Cell Design Specification — T8-40g

VBF-T8R1-DS-01 | Case t8_r1 — Long-endurance drone battery | 2026-08-25

Prepared: ________ Reviewed: ________ Approved: ________

Cell Design Specification — T8-40g Long-Endurance Drone Battery

- **Document code**: VBF-T8R1-DS-01
- **Case**: t8_r1 — long-endurance drone battery (energy density ≥ 446.18 Wh/kg, 5C discharge with capacity retention ≥ 90%, cell mass ≤ 40 g)
- **Generation date**: 2026-08-25
- **Signature**: Prepared: ________ Reviewed: ________ Approved: ________

All values below are mechanically taken from tool output files under `cell/` (PyBaMM DFN simulations, `calc-energy`) or from the Chen2020 base parameter set probe (`cell/chen2020_base_probe.json`). Values marked "literature/estimate" are annotated as such.

1. Basic Specification

Item	Value	Source
Electrochemical system	NMC811 / graphite (Chen2020 baseline teaching parameterization)	`--base Chen2020`
Nominal capacity	6.1494 Ah	`r5_final_params.json` (Chen2020 5.0 Ah × area scaling 1.2299)
Verified 1C capacity	6.9427 Ah	`r5_final_1c_dfn.json`
Voltage window	2.5 – 4.2 V	`chen2020_base_probe.json`
Discharge midpoint voltage	3.876 V	`r5_final_energy.json` (mechanical: time-midpoint voltage of 1C discharge)
Cell dimensions (unwound electrode geometry)	height 72.085 mm × width 1752.2 mm × thickness 0.1946 mm	`r5_final_params.json`, `r5_final_energy.json` (Σ layer thickness)
Shell / packaging thickness	Not provided (no parameter in set)	—
Electrolyte formulation	LiPF6 in organic carbonate (Chen2020 baseline; set defines transport properties, no explicit solvent-composition key)	probe
Transport overrides	conductivity σ = 1.3 S/m (constant), cation transference number t■ = 0.5, diffusivity D = 3.5×10■¹■ m²/s (constant)	`r5_final_params.json`; literature-class estimates (annotated in evaluate log)
Electrolyte additive candidates	Not proposed (start_stage = 3; no Stage-2 additive screening)	funnel entry
Cation transference number	0.5 (override; Chen2020 base 0.2594)	`r5_final_params.json`, probe

2. Electrode and Separator

Layer	Thickness (μm)	Porosity	Density (kg/m³)	Areal mass (kg/m²)	Source
Positive electrode (NMC811)	75.6 (base)	0.45	3262	0.13563	`r5_final_params.json`, probe, `r5_final_energy.json`
Negative electrode (graphite)	95.0 (override; base 85.2)	0.35	1657	0.10232	same
Separator	10.0 (override; base 12.0)	0.55	397	0.00179	same
Positive current collector (Al)	8.0 (override; base 16.0)	—	2700	0.02160	same
Negative current collector (Cu)	6.0 (override; base 12.0)	—	8960	0.05376	same
Total stack thickness	194.6 μm	—	—	0.31510	`r5_final_energy.json`

Particle radii: positive 1.0 μm (base 5.22 μm), negative 1.5 μm (base 5.86 μm) — `r5_final_params.json`, probe.

N/P ratio = negative capacity density × thickness ÷ positive capacity density × thickness = **0.96** (assembled-lithium basis, formula in `calc.xlsx` sheet "N/P and mass"): capacity density = (1–ε) × c_max × F/3600 × usable window, with windows taken from the only stoich data the set provides (initial concentrations: x■ = 0.270 positive, y■ = 0.9014 negative). Chen2020 has no stoich-limit keys (probe `_stoich_keys: []`). Full-theoretical-window (Δx=Δy=1) ratio = 0.78, also shown in `calc.xlsx`.

3. Process Design Parameters

Parameter	Formula	Value	Unit	Source
Positive areal density	thickness × (1–ε) × density	135.63	g/m²	`r5_final_energy.json` layer_kg_m2
Negative areal density	thickness × (1–ε) × density	102.32	g/m²	same
Positive compaction density	density × (1–ε) ÷ 1000	1.794	g/cm³	probe (3262 × 0.55)
Negative compaction density	density × (1–ε) ÷ 1000	1.077	g/cm³	probe (1657 × 0.65)
Electrolyte fill amount	pore volume × 1.2 g/cm³ (literature value, fill factor 100%)	9.19 cm³ → 11.03 g	g	pore volume from layer porosities × area; density literature-annotated
Formation recommendation	0.1C CC to 4.2 V, 25 °C, 2 cycles (design recommended value; production value requires tuning)	—	—	annotated recommendation

4. Mass Breakdown (contract caliber, electrolyte excluded)

Component	Mass (g)	Source
Positive electrode layer	17.131	`r5_final_energy.json` layer_kg_m2 × area
Negative electrode layer	12.923	same
Separator	0.226	same
Al current collector	2.728	same
Cu current collector	6.790	same
Total (contract caliber)	**39.798**	same (`mass_kg` 0.0398)
Electrolyte (literature density, NOT in contract mass)	11.03	pore volume × 1.2 g/cm³, annotated

Contract formula: ED = ∫V·I■C dt ÷ Σ layer_thickness × (1–porosity) × density × area. Electrolyte excluded (parameter set has no electrolyte density) — annotated honestly.

5. Performance Verification (vs entry-0 criteria)

Item	Value	Criterion	Verdict	Source
Energy density	634.32 Wh/kg	≥ 446.18 Wh/kg	✓ PASS	`r5_final_energy.json`
Energy density (volumetric)	1027.10 Wh/L	— (informational)	—	same
1C capacity	6.9427 Ah, T_max 299.76 K	—	✓	`r5_final_1c_dfn.json`
5C discharge capacity / retention	6.8571 Ah / 98.77%	≥ 90%	✓ PASS	`r5_final_5c_dfn.json`, `r5_final_retention.json`
5C discharge T_max	319.99 K	— (informational)	✓	`r5_final_5c_dfn.json`
Cell mass	39.800 g	≤ 40 g	✓ PASS	`r5_final_energy.json`
4C charge @45 °C T_max	330.41 K	≤ 333.15 K	✓ PASS (margin 2.74 K)	`r5_final_4c45_dfn.json`
Lithium plating (4C @45 °C)	anode potential min +0.0082 V > 0	no plating	✓ PASS	`r5_final_4c45_dfn.json` (auto-derived: min(anode_potential_v) ≥ 0)
DCR / power density	0.684 mΩ / 156.1 kW/kg	— (informational)	—	`r5_final_energy.json`

Honest caveat: 4C charge at 45 °C reaches the 4.2 V upper cut-off after only 0.9395 Ah charged (voltage-limited acceptance at 4C); fast charge at 4C is therefore thermally safe and plating-free but capacity-limited. Charging at lower C-rate removes the voltage limitation.

6. Design Notes (change history vs baseline, citing evaluate log)

Change	Baseline (Chen2020)	Final	Why (evaluate log round)
Positive/negative particle radius	5.22 / 5.86 µm	1.0 / 1.5 µm	R2 V4: baseline 5C retention was 8.7% — solid-state diffusion time constant $\tau = r^2/D_s$ limits 5C delivery; smaller particles raise rate capability
Electrolyte conductivity / t■ / diffusivity	conc.-dependent func / 0.2594 / func	1.3 S/m const / 0.5 / 3.5e-10 const	R2 V4: transport upgrade reduces electrolyte polarization at 5C (literature-class estimates)
Electrode porosity	0.335 / 0.25 / 0.47	0.45 / 0.35 / 0.55	R2 V4: higher porosity improves ion transport at 5C (costs ED, accepted — ED margin large)
Separator thickness	12 µm	10 µm	R2 V2/V4: thinner separator cuts ionic path resistance
Negative electrode thickness	85.2 µm	95.0 µm	R3 V7: thickens anode to add lithiation-capacity headroom (N/P↑, plating margin +10× at 4C)
Current collector thickness	16 / 12 µm	8 / 6 µm	R2 V2: thin foils raise gravimetric ED (ceiling assessment: 487 Wh/kg reachable in-architecture)
Cooling coefficient h	10 W/m^2/K	40 W/m^2/K	R3 V5 (30) → R4 V8 (35) → R4 V9 (40): 4C-45 °C exam T_{max} 354.3 K → 330.41 K (≤ 333.15 K); forced-air cooling assumed
Geometry / capacity scaling	0.065 m × 1.58 m, 5.0 Ah	0.072085 m × 1.7522 m, 6.1494 Ah	R5: area scaling ×1.2299 to bring mass to 39.8 g (≤ 40 g); verified scaling-invariant (ED/retention/T_{max} identical)

No system switch: ceiling assessment (R1) showed the 446.18 Wh/kg target is reachable within the Chen2020 architecture (thin-foil ceiling ≈ 487 Wh/kg), so no Stage-2 escalation was triggered.

7. Safety Determination

Item	Value	Criterion	Verdict	Source
T_{max}, 4C charge @45 °C	330.41 K	≤ 333.15 K	✓ PASS	`r5_final_4c45_dfn.json`
Plating, 4C charge @45 °C	anode min +0.0082 V	≥ 0 V	✓ PASS	same
Overcharge / nail / crush / drop	Not simulated	—	N/A	protocol has no such task criteria; see DVPR

Safety thresholds (333.15 K, plating) are protocol defaults — task text is silent on safety; recorded in entry 0 meta.